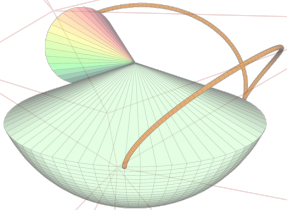

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Business Intelligence and Information Exploitation

Business Intelligence and Data Visualization

Professional Core

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Introduction

Information

- As a sales manager, imagine predicting each customer's spending and product choices for the next six months.
- As a productivity manager, imagine identifying which production teams create the highest quality products.
- As a broker negotiating electricity contracts, imagine being able to accurately predict the region's electricity demand for the next 90 days.
- As a supply chain manager, imagine determining which vendors provide the highest quality products the fastest.
- As a customer retention manager at a credit card company, imagine identifying customers likely to cancel their cards a month in advance.
- These are just a few examples of the kind of knowledge that can be exposed and exploited through the use of Business Intelligence (BI).

Introduction

Information

- In the last few years, the ability to create, collect, and store information has widely outpaced our ability to make significant use of that information.
- Yet there is significant hidden value locked away in corporate databases, waiting to be discovered and exploited.
- But to expose this hidden value, we must first gain an understanding of how to train ourselves and our colleagues to think about information in a different way.
- Instead of viewing data like raw materials, think of it as a valuable corporate asset that can be leveraged in various ways for business advantage.

Why Business Intelligence

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Why Business Intelligence

1. Increased Profitability

- Business intelligence assists business clients in evaluating customer lifetime value and short-term profitability, enabling differentiation between profitable and non-profitable customers.
- For instance, financial consultants note that in a typical retail bank portfolio, 20% of accounts contribute profits totaling 200% of the overall return, while over half of the accounts incur losses.

2. Decreased Costs

- BI can evaluate organizational costs, including improved logistics management, reduced operational expenses like warehousing and delivery costs, and optimizing sales investments.

Why Business Intelligence

3. Improved CRM

- Customer Relationship Management (CRM) leverages BI to analyze aggregated customer data, enhancing customer service responsiveness, uncovering cross-sell and up-sell opportunities, and fostering greater customer loyalty.

4. Decreased Risk

- Applying business intelligence methods to credit data enhances the accuracy of credit risk analysis, while comprehensive analysis of supplier and consumer activity provides insights into optimizing and streamlining the supply chain operations.

Why Business Intelligence

- There are many other benefits to building a BI practice within an organization. Some of those focus on the most basic questions about how a company does business.
  1. How many customers do you have?
  2. For each product, how many were sold over the last 12 months?
  3. Who are your 20 best customers?
  4. What is the value of any particular customer?
  5. Who are your 20 best suppliers?
- BI practice should not only offer the tools to answer these questions but also serve as a catalyst for a transformative shift in business thinking and the strategic use of information.
- For example, To identify the top 20 customers, it's essential to define what constitutes a "good" versus a "bad" customer and establish comprehensive metrics for measurement.
- This involves collecting and integrating relevant data, ensuring data quality, and effectively communicating the results to support strategic decision-making.

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The Information Asset

- Most BI investment targets infrastructure, with little spent on effectively managing and utilizing company data.
- Lack of focus and clear success criteria often cause delays or failures in data warehousing, leading to unmet BI expectations.
- Some organizations recognize data as a corporate asset, understanding the value of proper collection, aggregation, and analysis.
- These efforts can enhance operational processes and offer deeper insights into customer profiles and behavior.
- In these environments, BI is clearly defined as a set of tools and methodologies aimed at extracting actionable knowledge from a company's information assets.

Exploiting Information

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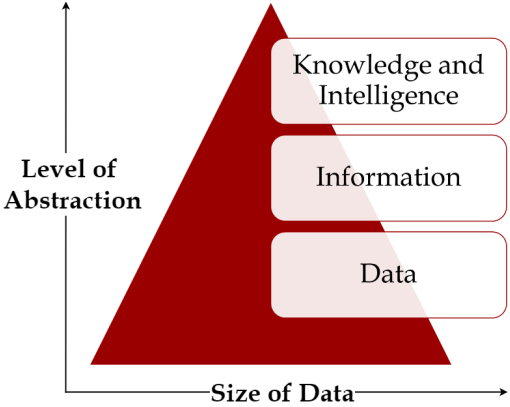
Exploiting Information

- BI professionals focus on leveraging information to enhance organizational operations.
- Effective data exploitation hinges on managing information flow into an intelligence platform structured by a pyramid of abstraction, ensuring optimal access to knowledge assets.
- Effective information exploitation relies on hierarchical evolution in information and knowledge management, each with distinct benefits and challenges.



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BI and Program Success

- The differing views on what BI is and how to use the knowledge it provides often result in project failures.
- Companies starting data warehousing projects or buying data mining software often have high expectations but face many disappointments.
- These failures are often due to issues in the conception, design, architecture, management, and implementation of data warehouse projects.
- The unclear understanding of BI methods and products led to the lack of a clear value proposition from the business sponsor.
- The scope of the project was not fully understood, causing delays in delivering to the business sponsor.

BI and Program Success

- Insufficient technical training prevented developers from getting software products to do what the vendors said they do.
- Poor understanding of technology infrastructure led to poor planning and scheduling.
- Business users were unable to trust results due to poor data quality.
- The lack of a clear statement of success criteria, along with a lack of ways to measure program success, led to a perception of failure.
- It also highlights key issues such as:
  1. Management issues
  2. Technical issues
  3. Performance issues
  4. Complexity

What is BI?

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What is BI?

> *" The processes, technologies, and tools needed to turn data into information, information into knowledge, and knowledge into plans that drive profitable business action. Business intelligence encompasses data warehousing, business analytic tools, and content/knowledge management."*

- A BI practice is more than just a collection of tools. This means that without the processes and the right people, the tools are of little value.
- The value of BI is realized in the context of profitable business action. This means that if knowledge that can be used for profitable action is ignored, the practice is of little value.

Data

- Data is a collection of raw value elements or facts used for calculating, reasoning, or measuring.
- Data may be collected, stored, or processed but not put into a context from which any meaning can be inferred.

What is BI?

Information

- Information is the result of collecting and organizing data in way that establishes relationships between data items, which thereby provides context and meaning.

Knowledge

- Knowledge is the concept of understanding information based on recognized patterns in a way that provides insight to information.

Turning Data into Information

- Turning data into information involves deciding what data to collect and manage and in what context.
- A good example is designing a database that models real-world entities, like people and organizations, and their roles.

What is BI?

Turning Data into Information

- Individual bits of data, like names and birth dates, have little value out of context. By defining and organizing this data within a description of a party, we create context and turn data into information.
- This aspect of BI involves managing and presenting data through hardware, database systems, and software tools.
- This aspect includes tools for querying and reporting to access the data effectively.
- Expertise in data management is crucial for integrating and coordinating this technology effectively.

Turning Information into Knowledge

- It's that sudden flash of insight that magically provides the answer to a tough problem you've been agonizing over.
- It can happen when you wake up at night, in traffic, or daydreaming in the shower.

What is BI?

Turning Information into Knowledge

- Turning information into knowledge is like experiencing a sudden flash of insight.
- Piles of information are analyzed in various ways until critical knowledge emerges.
- Critical knowledge is valuable because it helps create actionable plans to solve business problems.
- This aspect of BI includes analytical components like data warehousing, OLAP, data quality, data profiling, business rule analysis, and data mining.
- Having tools for these functions is useless without professionals who know how to use them effectively and achieve the right results.

Turning Knowledge into Actionable Plans

- The last aspect is the most important, as it is where real value is derived.
- Using BI for micro-marketing is pointless if you don't have a plan to contact the right customer.

Turning Knowledge into Actionable Plans

- Using BI for fraud detection is of little value if your organization doesn't take steps to prevent the fraudulent behavior.
- The key to any BI strategy is taking action based on the intelligence gathered.
- These actions allow senior management to see the true return on investment for their IT spending.
- A BI program boosts business efficiency, increases sales, improves customer targeting, reduces service costs, detects fraud, and enhances profits while cutting costs.
- When implemented properly, BI stands out as an IT area that can generate profits instead of being seen solely as a traditional cost center.

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Actionable Knowledge

- Knowledge is only valuable if it results in value-producing actions. Without actionable outcomes, discovered knowledge remains of little worth.
- The business and technology teams must collaborate to act quickly on discovered intelligence to maximize benefits.
- This highlights that BI success depends on cooperation between technical developers and business clients.

Books & E-Resources

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Books & E-Resources

Textbooks

1. *Loshin, D. (2012). Business intelligence: the savvy manager's guide. Newnes.*

Reference Books & E-Resources

1. *Kemper, H. G., Mehanna, W., & Unger, C. (2010). Business Intelligence-Grundlagen und praktische Anwendungen (Vol. 3). Wiesbaden: Vieweg+ Teubner.*
2. *Moss, L. T., & Atre, S. (2003). Business intelligence roadmap: the complete project lifecycle for decision-support applications. Addison-Wesley Professional.*
3. *Business Intelligence Concepts, tools, and applications. (2021, October 9). Coursera.* `https://www.coursera.org/learn/business-intelligence-tools`

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Brainstorming

Question: Conduct an in-depth study on implementing Business Intelligence in a clothing store. Evaluate the program's success, identify the types of data collected, classify the data into information and knowledge, propose actionable and non-actionable insights that are practical and applicable, analyze roadblocks faced during implementation, and elucidate how Business Intelligence transforms raw data into actionable knowledge for strategic decision-making in the clothing store. Provide real-world examples, also design a database schema for it.

Note: Use AI tools to generate your answers. Understand and write answer on your ALM book.

ALM - I: Brainstorming

Rubrics

10M	7M - 9M	4M - 6M	1M - 3M
Able to explain the topic and offer strong views	Able to explain the topic and offer good views	Able to define the topic and offer fair views	Weak in conveying or couldn't express at all.

AI Tools

ChatGPT: For finding solution with example.
`https://chatgpt.com/`

Google Gemini: For finding solution with example.
`https://gemini.google.com/?hl=en-IN`

DiagramGPT: For generating database schema.
`https://www.eraser.io/diagramgpt`

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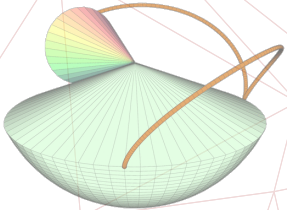

The end!